# Transitivity, potency and cumulation hold for every stage V_alpha of the cumulative hierarchy

The Open Logic Project contributors

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the simultaneous induction framing, all three successor-case verifications and the limit-case insertion argument are verbatim; the closing union observation for the two remaining limit properties is a standard closure fact stated in full sentence.

*Editorial note.* The label Valphabasicprops names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lem environment in the unit source.

*Allowed standard prerequisites (editorial).*

- Successor and limit case division for ordinals. Ordinals unit background: every ordinal is empty, a successor, or a limit. The three cases of the simultaneous transfinite induction at lines 862-882.
- Definition of the hierarchy V_alpha. Same unit, the cumulative hierarchy definition preceding this lemma: V_empty, V_succ beta = Pow(V_beta), V_limit = union. Every unfolding of V_alpha in the proof runs through these equations.

**Lemma 0.1.** *For each ordinal $\alpha$:*

1. *Each $V_\alpha$ is transitive.*
2. *Each $V_\alpha$ is potent.*
3. *If $\gamma \in \alpha$, then $V_\gamma \in V_\alpha$ (and hence also $V_\gamma \subseteq V_\alpha$ by 1)*

*Proof.* We prove this by a (simultaneous) transfinite induction. For induction, suppose that 1–3 holds for each ordinal $\beta < \alpha$.

The case of $\alpha = \emptyset$ is trivial.

Suppose $\alpha = +1\beta$. To show 3, if $\gamma \in \alpha$ then $V_\gamma \subseteq V_\beta$ by hypothesis, so $V_\gamma \in \mathcal{P}V_\beta = V_\alpha$. To show 2, suppose $A \subseteq B \in V_\alpha$ i.e., $A \subseteq B \subseteq V_\beta$; then

$A \subseteq V_\beta$ so $A \in V_\alpha$. To show 1, note that if $x \in A \in V_\alpha$ we have $A \subseteq V_\beta$, so $x \in V_\beta$, so $x \subseteq V_\beta$ as $V_\beta$ is transitive by hypothesis, and so $x \in V_\alpha$.

Suppose $\alpha$ is a limit ordinal. To show 3, if $\gamma \in \alpha$ then $\gamma \in +1\gamma \in \alpha$, so that $V_\gamma \in V_{+1\gamma}$ by assumption, hence $V_\gamma \in \bigcup_{\beta\in\alpha} V_\beta = V_\alpha$. To show 1 and 2, just observe that a union of transitive (respectively, potent) sets is transitive (respectively, potent). $\square$